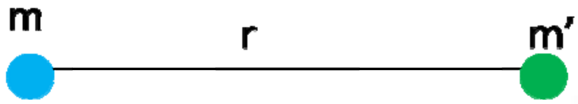
Gravitation:

Every particle attracts other particles towards it with a force known as gravitational Force.



Let us consider two point masses m & m' separated by a distance r.

Let F = Gravitational force of attraction between the point masses.

According to Newton:

$$F \propto mm' \longrightarrow (1)$$

$$F \propto \frac{1}{r^2} \longrightarrow (2)$$

$$F = G\frac{mm'}{r^2}$$

Where G is constant of proportionality and is known as Gravitational Constant.

$$[G] = \frac{[F][r^2]}{[m][m]} = \frac{MLT^{-2}L^2}{MM} = [M^{-1}L^3T^{-2}]$$

$$Unit = gm^{-1}cm^3\sec^{-2}, kg^{-1}m^3\sec^{-2}$$

The value of G =6.667 x 10^{-11} $kg^{-1}m^3\sec^{-2}$